



General histology

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College of Medicine
Department: Microbiology .,
Physiology & Genetics
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Learning outcome:

- Student must know structure, function and characteristic feature of nerve cells (neurons).
- Student must know different categories for classification of neurons.
- Student must know structure of synapse & how the nerve impulse will transmit from one neuron to another.

Nervous Tissue

- Nervous system = communication system of body formed by a network of more than 100 million nerve cells (neurons), assisted by many more glial cells.
- Anatomically, it divided into 2 parts:
 - *Central Nervous System (CNS)* = brain and spinal cord.
 - *Peripheral Nervous System (PNS)* = spinal and cranial nerves and ganglia.

properties:

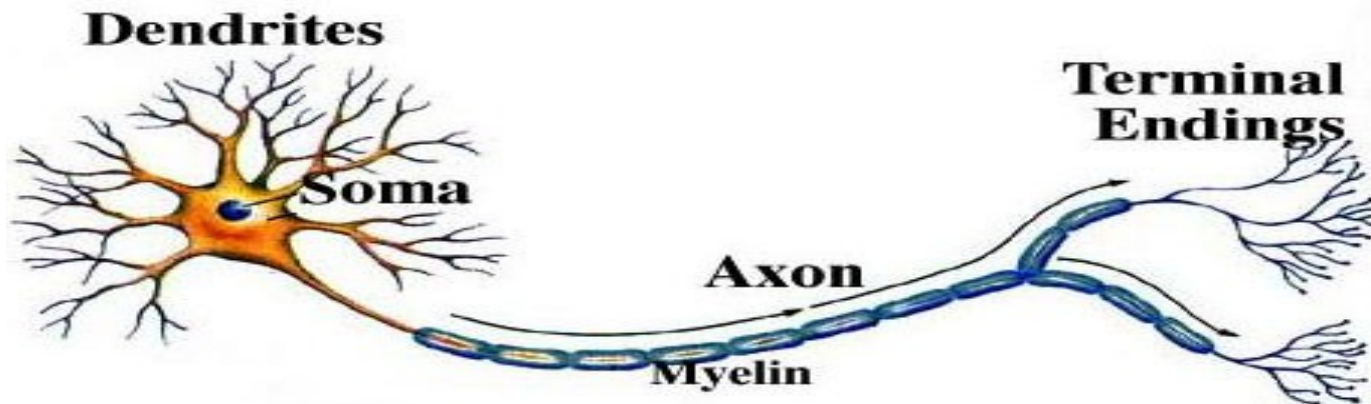
- **a- irritability** (ability to react to various kinds of stimuli , physical, chemical , internal or external: excitation),
- **b- conductivity** (nerve impulses): ability to transmit these excitations in the form of nerve impulses from one locality to another.

Histologically nervous system is composed of

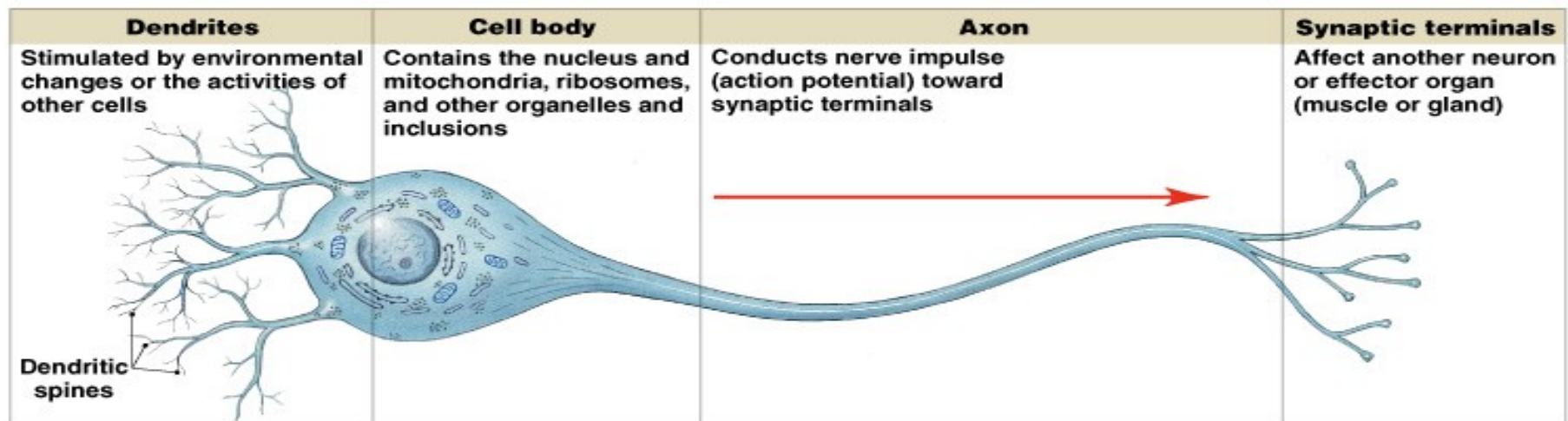
- Cells = Neuron (nerve cell)& glial cells
- Neurons are Consists of:
 - Cell Body (perikaryon)- contains nucleus
 - *Dendrites* = stimulus receiving and impulse-generating component, carries impulse to cell body
 - *Axon* = propagates impulse, carries impulse away from cell body.

The Neurons

Basic structural and functional unit of the nervous system
Specialized to conduct information from one part of the body to another,
consisting of the cell body (perikaryon, soma) and
processes of two types= dendrites and axon.

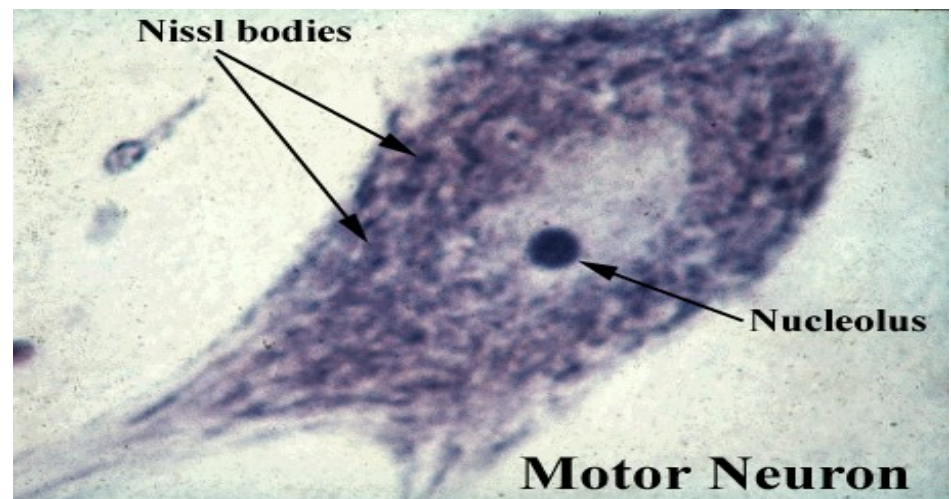
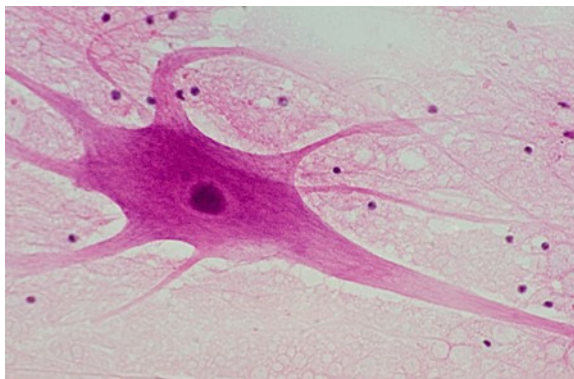


Neuron structure



The Neurons

A- nucleus: lies near center of cell body and is pale-staining (large spherical euchromatic pale) with dense single dark-staining nucleolus,

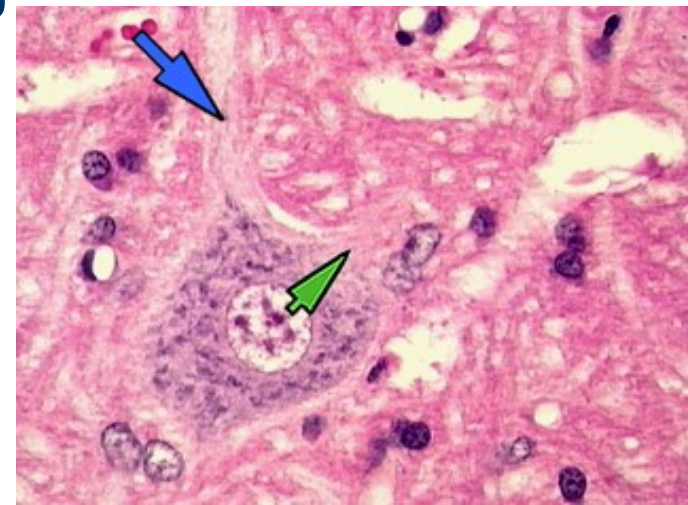


B- cytoplasm:

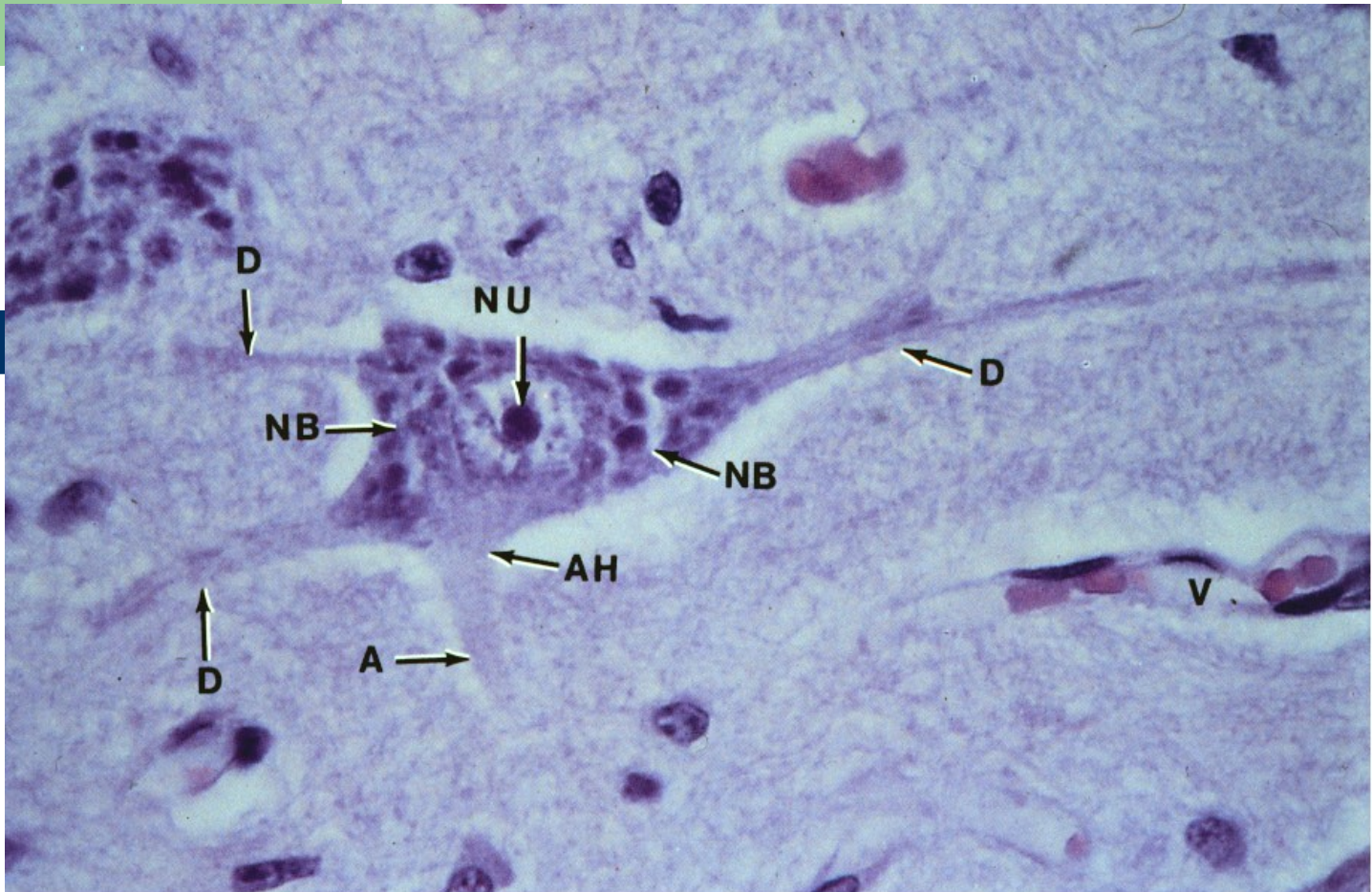
The protein synthesizing components extensive rough **RER** known as Nissl bodies. present except in the axon hillock.

Golgi complex located only in cell body around nucleus.

Mitochondria in perikaryon and also abundant in axon terminals.



- **Neurofilaments** are the cytoskeletal elements of neurons, parallel fine filaments, they aggregate and seen as fibrils by light microscope.
- **Microtubules** found in soma & processes- for transport.
- **Inclusion bodies, melanin, lipofuchsin & lipid droplets, glycogen.** (Increased with age).



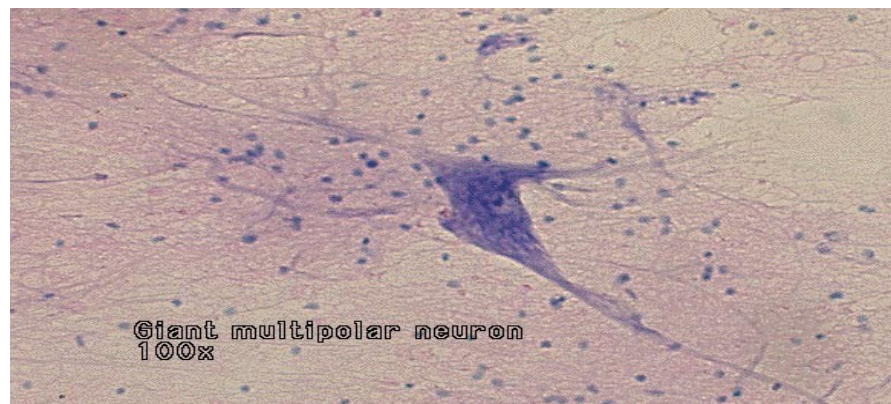
A-axon D-dendrite N-nucleus NB-
Nissl body
AH-axon hillock V-blood vessel NU-nucleolus

Dendrites

Short divide like branches of a tree except in case of **Purkinje neurons** where they divide in **one plane**. becomes thinner as they branch.

their main function is to receive incoming signals.

They effectively increase the surface area of a neuron to increase its ability to communicate with other neurons.

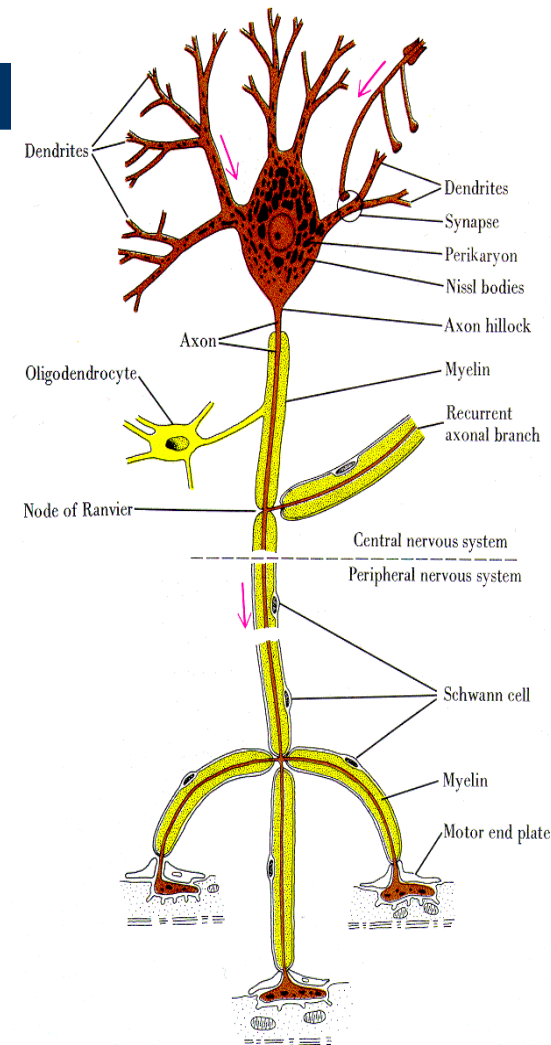


structure:

*Similar to soma but lack golgi.

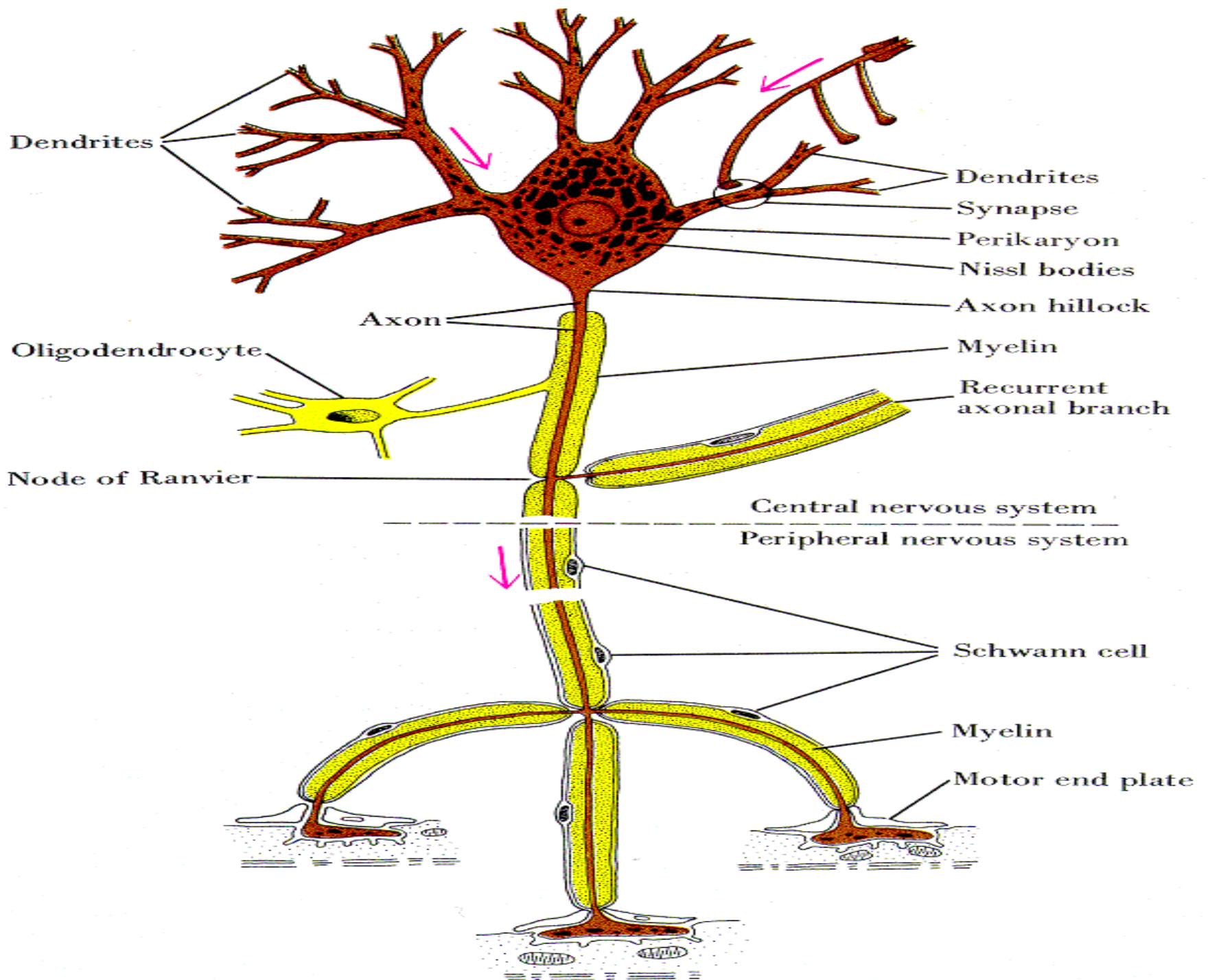
*Have short projections, spines

(**gemmules**) at sites of synaptic contact.



Axon

- One cylindrical process are thinner and longer than dendrites constant in diameter vary in length **designed to convey information away from the cell body** have **collateral branches**
- the area of origin of the axon soma is called **axon hillock**, devoid of nissl, pale-staining region;
- contains mitochondria but no ribosomes & RER,
- while the first part of it is called initial segment which is not myelinated.



Axons usually branch at their terminals called **terminal arborization** where there is a slight expansion called **end bulb- bouton terminal**, are the sites of synaptic contact.

Classification of neurons

a- according to the size ;

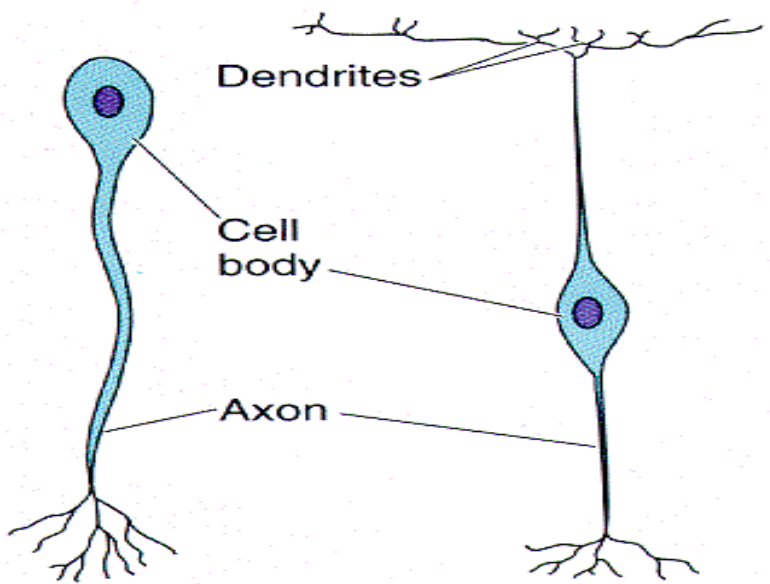
vary from very small 5 μm (granular neurons) up to very large 150 μm (motor neurons of spinal cord).

b-according to shape :

- **Spherical** (pseudounipolar in sensory ganglia),
 - **Pyriform ovoid** (purkinje neurons in the cerebellum)
- **Angular** like the stellate star shaped neurons of the spinal cord
- **pyramidal** neurons of cerebral cortex.

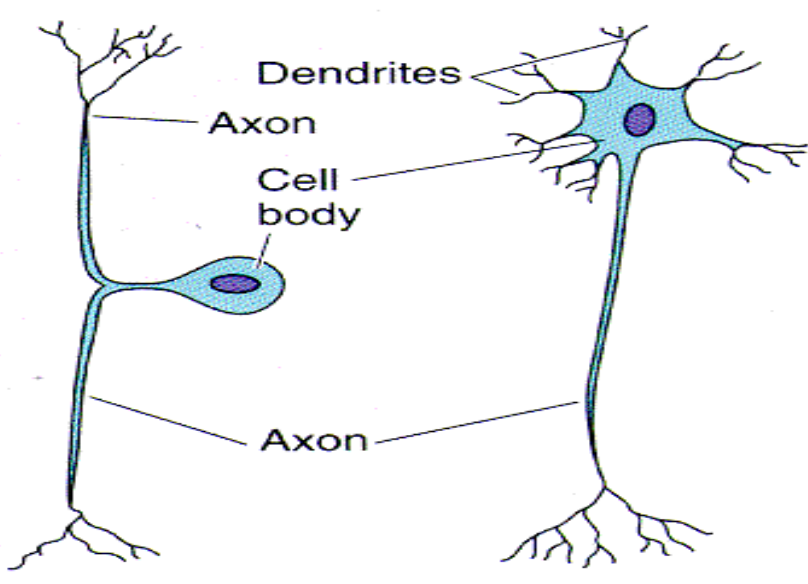
C-according to types:

- **Multipolar.** (most common, one axon and several dendrites) e.g. motor neuron in ventral horn of spinal cord)
- **Bipolar.** one axon & one dendrite found in olfactory epith, retina of the eye..
- **Unipolar.** cell body + axon only (photoreceptors of eye..)
- **(pseudounipolar).** which have single process that is close to the perikaryon & divided into 2 branches. eg. spinal ganglion & cranial ganglia



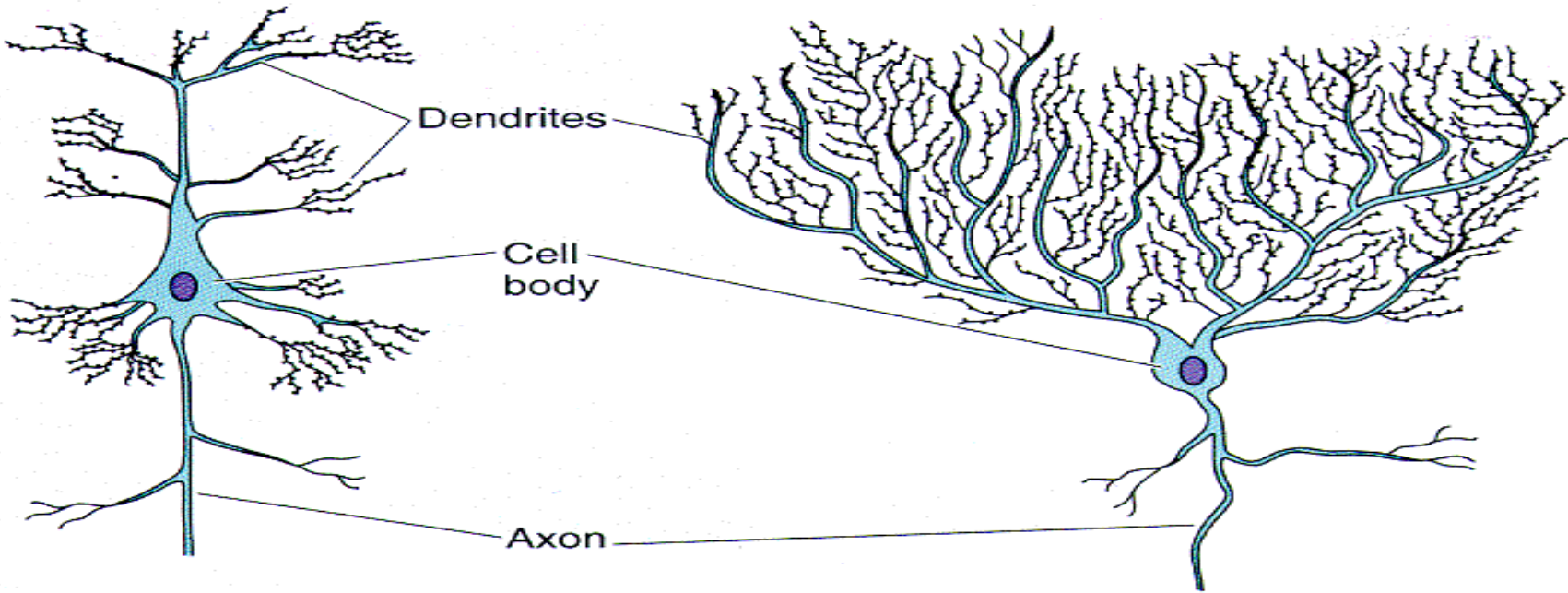
Unipolar

Bipolar
(retina)



Pseudo-unipolar
(dorsal root ganglion)

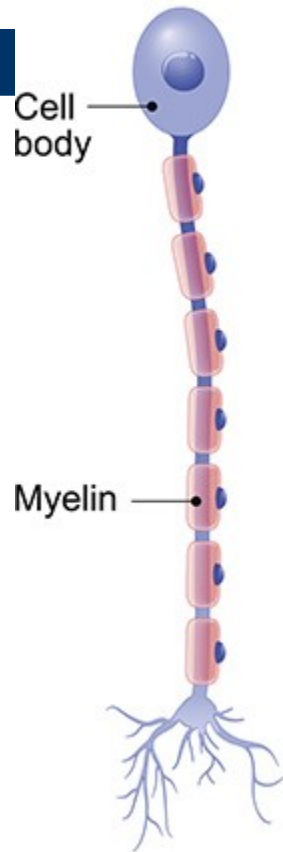
Multipolar
(motor)



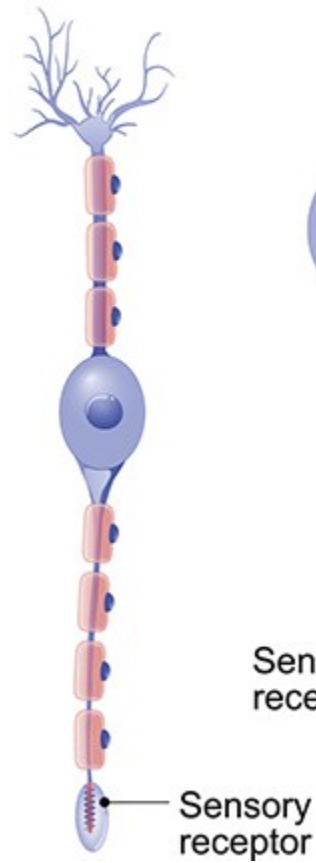
Pyramidal
(hippocampus)

Purkinje
(cerebellum)

Unipolar



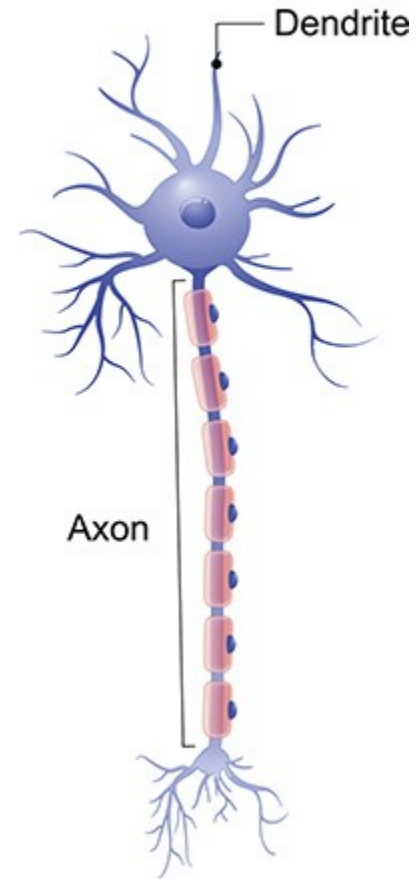
Bipolar



Pseudounipolar

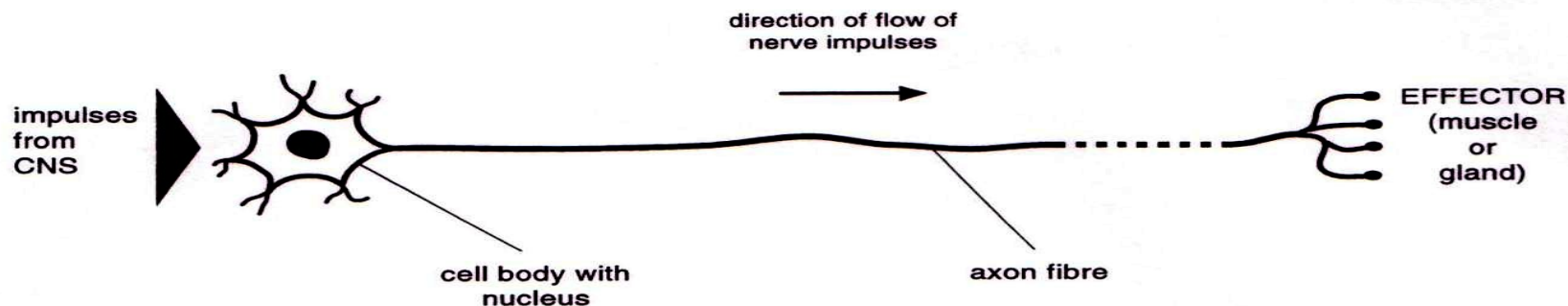


Multipolar



D- according to function:

- 1- **Motor neuron:** Carry impulses from cell body which lie in the central nervous system to effector cells.
control effector organs carry impulses to end organs induce or inhibit muscle contraction , glandular secretion e.g. multipolar neuron in spinal cord, Autonomic ganglia .



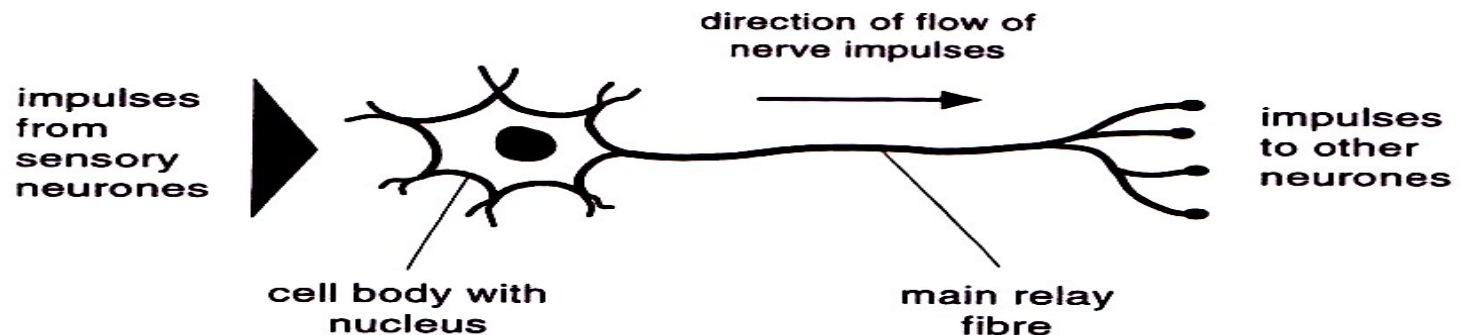
2- Sensory neuron

- Receive stimuli in & outside the body, receives the impulses generated by stimulation of peripheral sensory cells & organs , carry them toward CNS e.g. pseudo unipolar neuron in cranial and spinal ganglia

Carries impulses from receptors e.g pain receptors in skin to the CNS(brain or spinal cord)

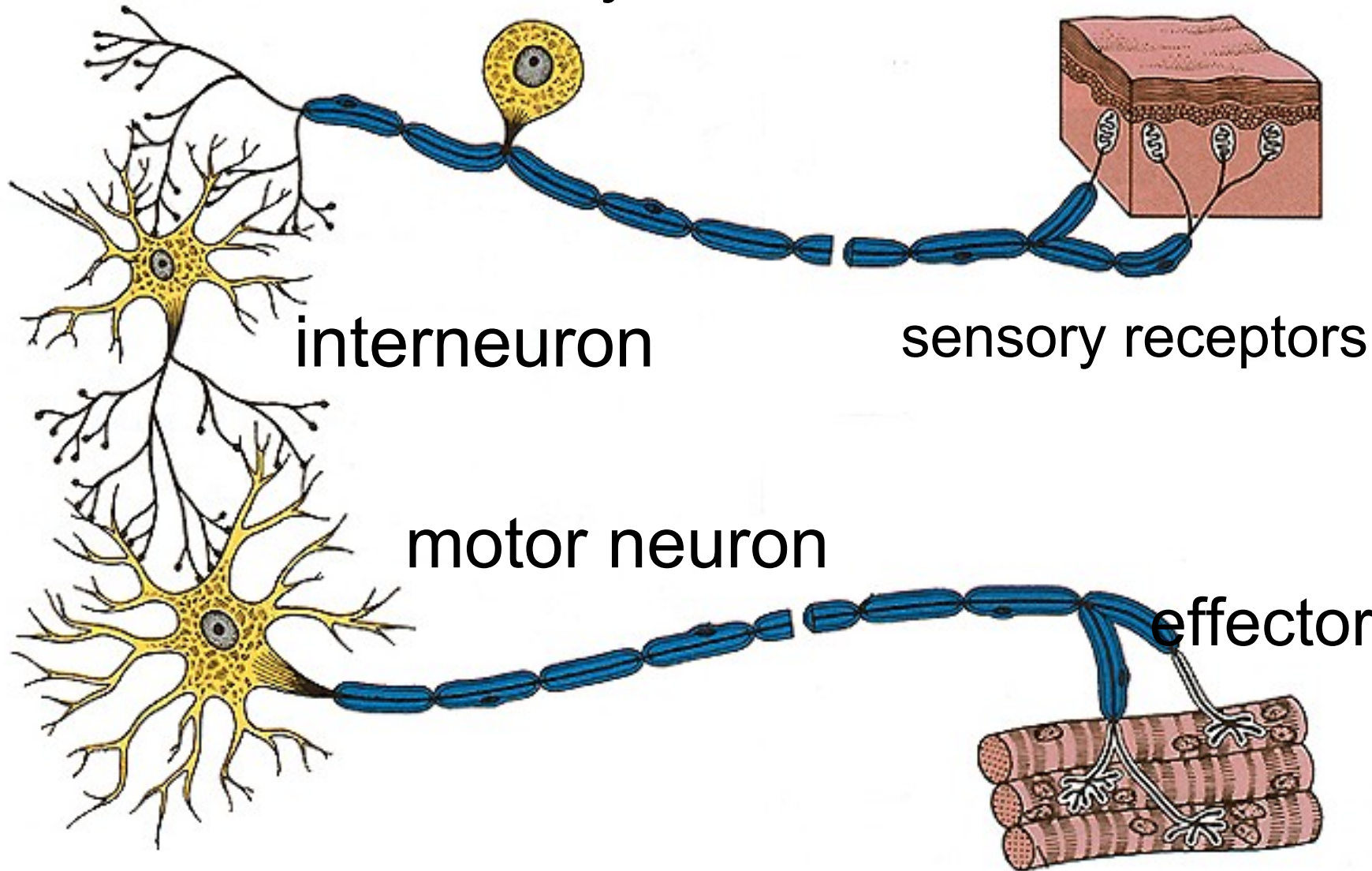
3- Interneuron:

Establish inter-relations with other neurons forming a complex chains or circuits, carry the signal between motor & sensory neuron.



*Carries impulses from sensory nerves to motor nerves.

sensory neuron



interneuron

sensory receptors

motor neuron

effector

sensory input

Skin

motor input

Receptor

Dendrite

Axon

Muscle

sensory receptor

effector

Axon bulb

Axon terminals

Axon

Sensory neuron

Cell body

Impulse direction

Motor neuron

Axon hillock

Dendrites

Interneuron

Cell body

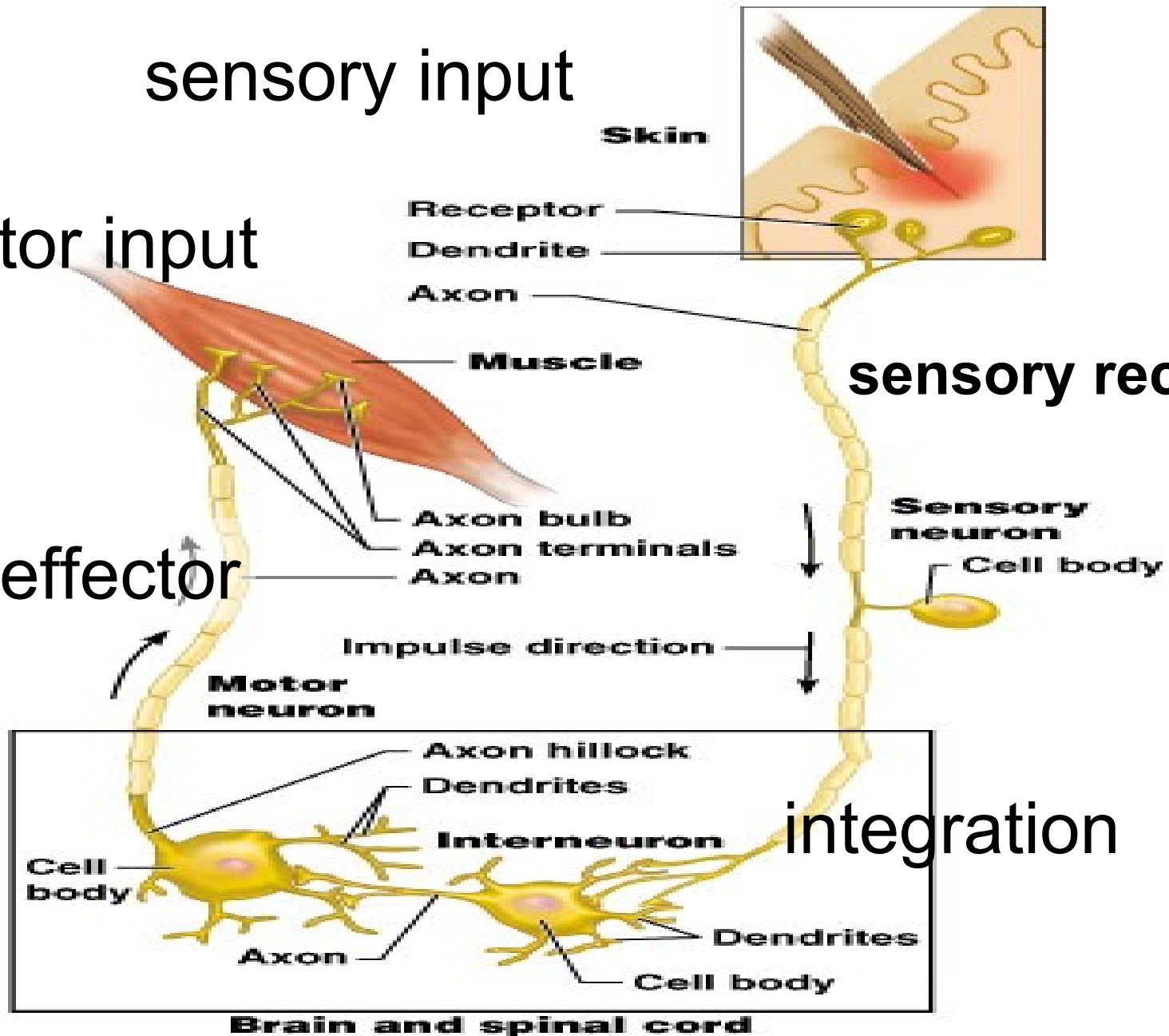
Axon

Dendrites

Cell body

integration

Brain and spinal cord



E- according to the length of processes:



Either very short axon do not exceed the gray matter.

e.g; Golgi type II.

Or very long up to 80 cm as in the motor neurons of the anterior Horn of spinal cord. is Golgi type I

Classification of neurons

a- according to the size

very small
5 μm (granular neurons)

very large
150 μm (motor neurons of spinal cord).

c- according to the length of processes

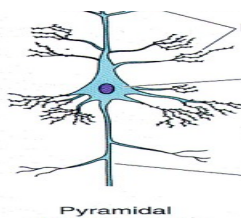
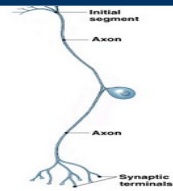
b-according to shape

1-Spherical
(pseudounipolar in sensory ganglia),

2-pyriform ovoid
(purkinje neurons in the cerebellum)

3-angular like the stellate
stare shaped neurons of the spinal cord

4-pyramidal neurons of cerebral cortex.



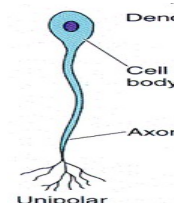
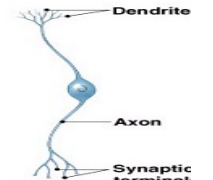
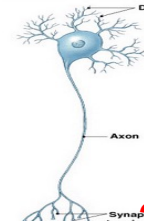
d-according to types

1-Multipolar
motor neuron in ventral horn of spinal cord

2-Bipolar
in olfactory epith, retina of the eye

3-Unipolar.
(photoreceptors of eye..)

4-(pseudounipolar)
spinal ganglion & cranial g. .

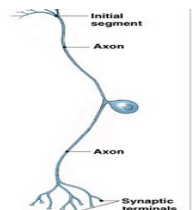


e- according to function

1- Motor

2- Sensory

3- Interneuron



The Synapse

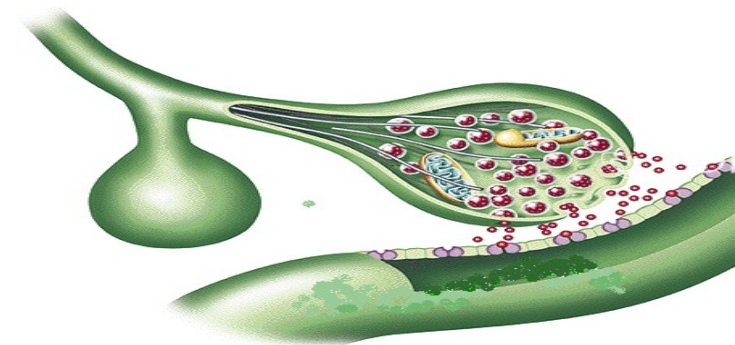
Are the sites of transmission of nerve impulses between neurons or between neurons & effectors organ. **Types:**

a-chemical (most common) where chemical sub. like **neurotransmitters** are used.

b- electrical (rare) electrical impulse is passed directly by **low resistance gap junctions**.

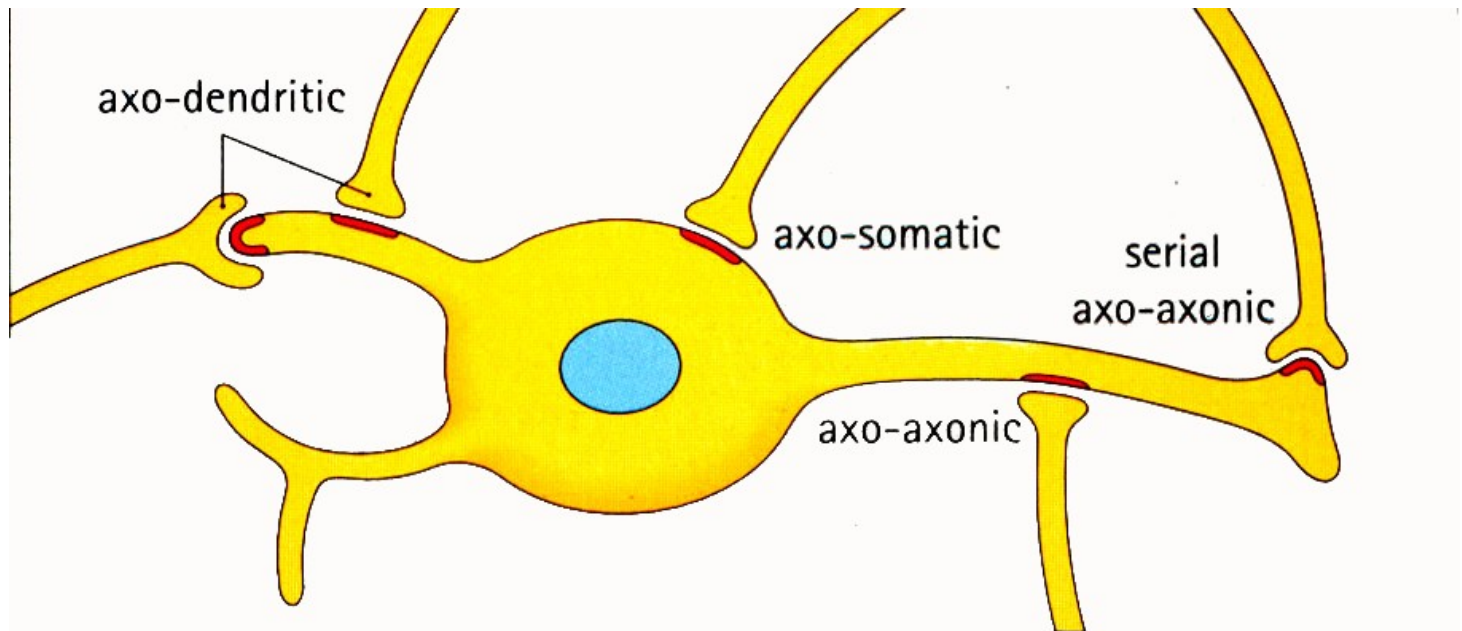
Functionally:

either **excitatory** or **inhibitory**.



Types of contact:

Axo-axonic. Axo-dendritic. Axo-somatic and dendro-dendritic

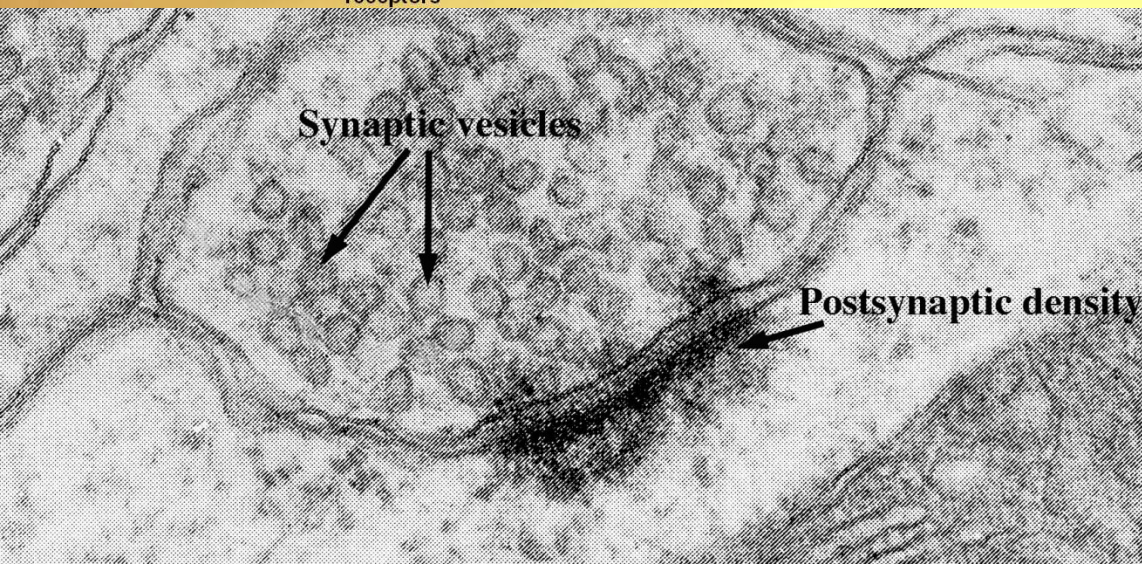
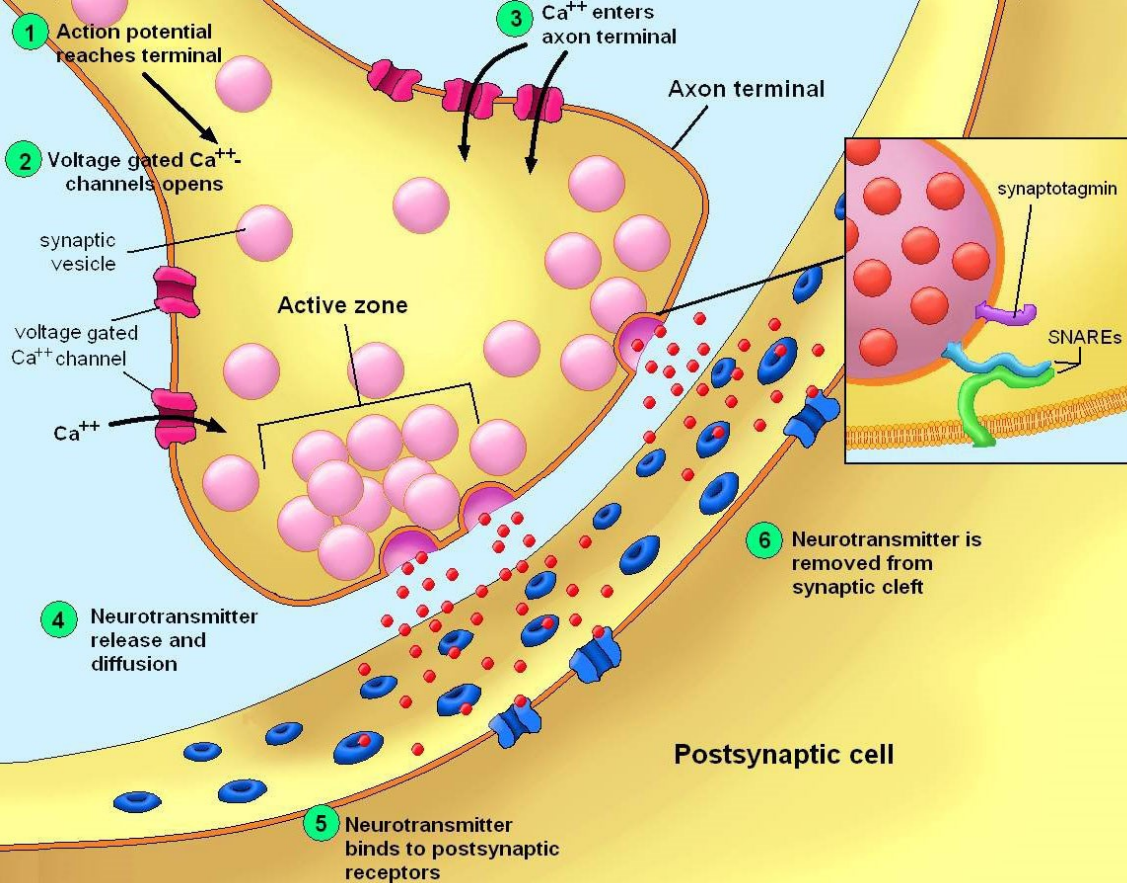


The Synapse

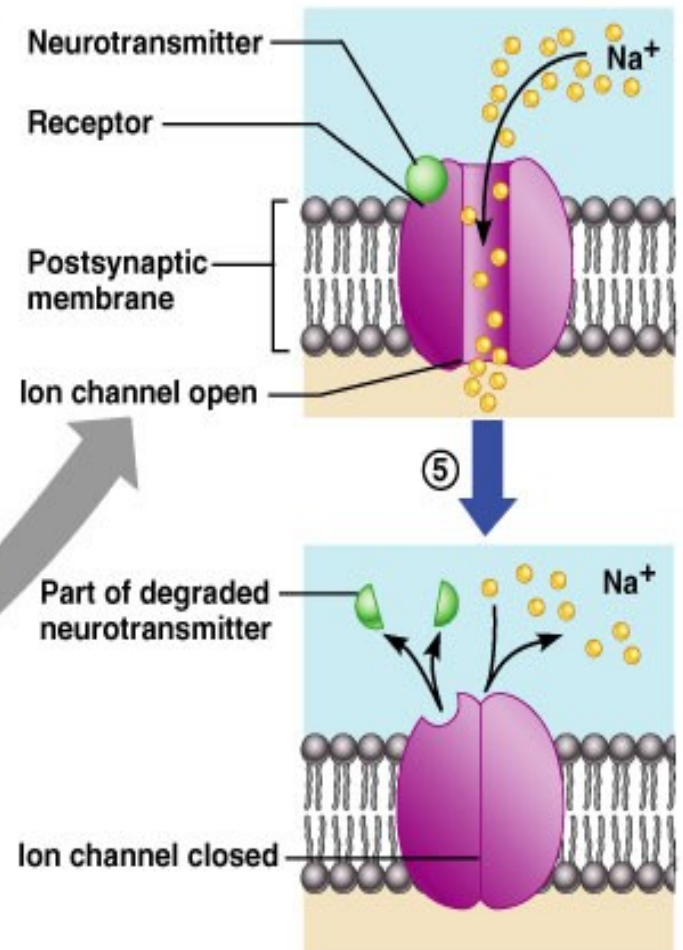
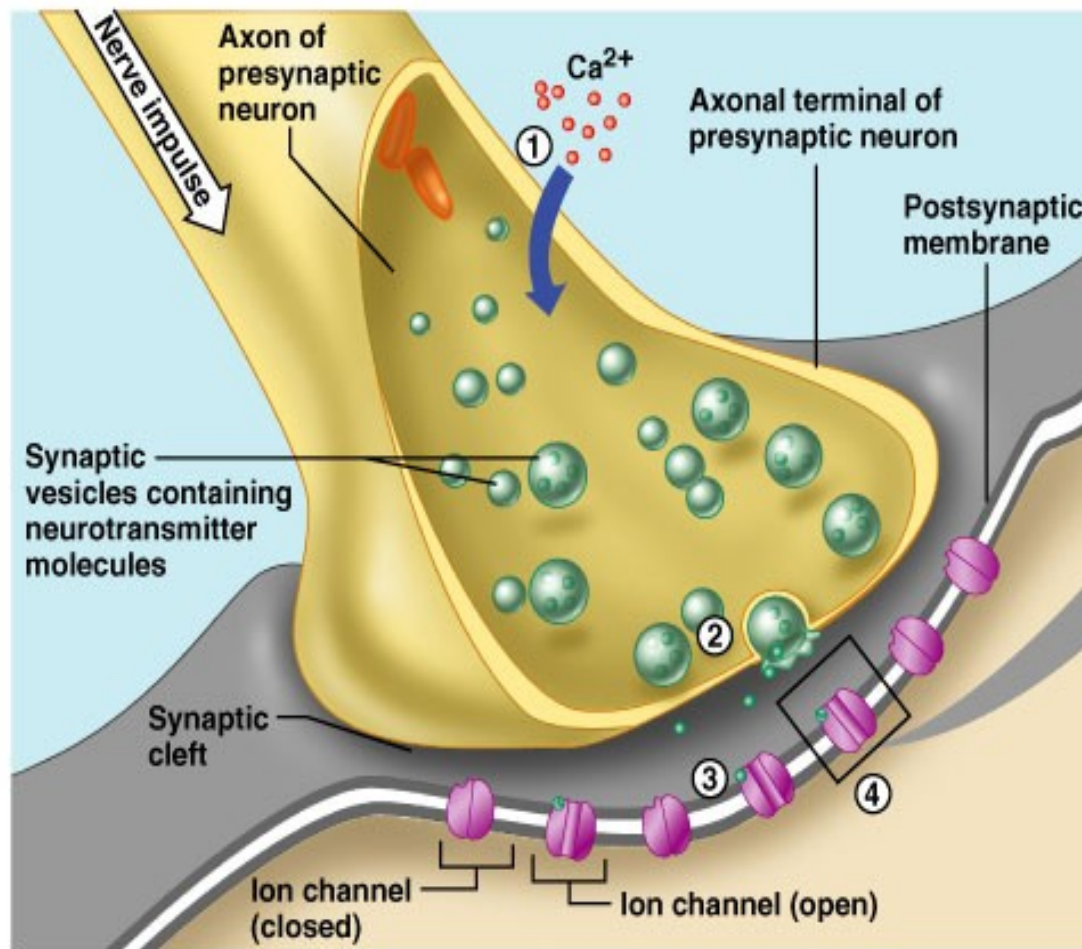
Structurally: three parts are recognized

- a- **pre-synaptic membrane:** of the end bulb (contains secretory vesicles contain neurotransmitters-mitochondria.
- b- **Synaptic cleft-gap:** fluid filled space between pre& post synaptic membrane.

c- post-synaptic membrane: this thickening of the plasma membrane of the next neuron or target cells e.g. muscle containing mitochondria but no secretory vesicles contain receptors for neurotransmitters .



At a chemical synapse neurotransmitter release is triggered by the influx of Ca^{2+} and postsynaptic neurotransmitter receptors receive the signal.



An example of a synapse:

The neuromuscular junction (motor endplate)

